Competition 01 - TechVenture Investment Challenge

Management Science



Client Briefing: TechVenture Innovation Fund

Company Background

TechVenture Innovation Fund is a venture capital firm specializing in early-stage technology investments. Founded in 2018, the fund has successfully backed 23 startups with an average return of 32% per year. The partners pride themselves on data-driven decision making and sophisticated risk analysis.

Your Role: You've been hired as consultants to analyze their latest investment opportunity.

The Investment Opportunity

TechVenture has **€2 million** available for immediate deployment. After extensive due diligence, they've narrowed their options to **four promising startups**. Due to diversification requirements and partnership agreements, they must invest in **exactly two startups**, allocating **€1 million to each**.

The Four Candidates

1. CloudAI Solutions

- **Industry:** Enterprise AI/SaaS
- **Product:** AI-powered business intelligence platform
- **Market Size:** €45B growing at 25% annually
- **Competition:** Moderate (established players but room for innovation)
- **Return Distribution:** Normal distribution
  - Mean return: 25% per year
  - Standard deviation: 15%
- **Key Risk:** Technology adoption speed, enterprise sales cycles

2. GreenGrid Energy

- **Industry:** Renewable Energy Technology
- **Product:** Next-generation solar panel storage systems
- **Market Size:** €30B growing at 18% annually
- **Competition:** High (many players, commoditization risk)
- **Return Distribution:** Normal distribution
  - Mean return: 18% per year
  - Standard deviation: 8%
- **Key Risk:** Regulatory changes, commodity price fluctuations

3. HealthTrack Pro

- **Industry:** Medical Devices / Wearables
- **Product:** FDA-approved continuous health monitoring wearable
- **Market Size:** €20B growing at 35% annually
- **Competition:** Low (first-mover in specific medical conditions)
- **Return Distribution:** Normal distribution
  - Mean return: 30% per year
  - Standard deviation: 25%
- **Key Risk:** Regulatory approval delays, clinical trial outcomes

4. FinFlow

- **Industry:** B2B Fintech
- **Product:** Automated payment reconciliation for enterprises
- **Market Size:** €15B growing at 20% annually
- **Competition:** Moderate (fragmented market)
- **Return Distribution:** Uniform distribution
  - Minimum return: 10% per year
  - Maximum return: 35% per year
- **Key Risk:** Customer acquisition cost, banking partnerships

Success Metrics

The fund evaluates investments based on multiple criteria:

1. **Expected Total Return:** The mean return across all scenarios
2. **Risk-Adjusted Return:** Return relative to volatility
3. **Downside Protection:** Probability of loss (return < 0%)
4. **Upside Potential:** Probability of exceptional returns (>50% total)
5. **Worst-Case Scenario:** Expected shortfall in bottom 10% of outcomes

Constraints and Requirements

- Must select exactly **2 startups**
- Must invest **€1 million in each** selected startup
- Cannot invest in more or fewer than 2 startups
- Investment horizon is **1 year** for this analysis
- Assume startups' returns are **independent** (no correlation)

The Challenge

Your Mission

Use Monte Carlo simulation to determine which pair of startups TechVenture should invest in. Your analysis should:

1. **Simulate** at least 10,000 scenarios for each startup
2. **Evaluate** all possible pairs (there are 6 combinations)

3. **Compare** portfolios using multiple risk metrics
4. **Recommend** the optimal investment pair
5. **Justify** your recommendation with data

Evaluation Criteria

Your presentation will be evaluated on:

- **Correctness (50%):** Accurate simulation and calculations
- **Business Reasoning (25%):** Clear justification aligned with fund goals
- **Presentation (25%):** Clear, professional one-slide summary

Time Limit

You have until **next session** to complete your analysis and prepare your presentation.

Data Access & Starter Code

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from itertools import combinations

# Set seed for consistency across teams (optional - you may change)
np.random.seed(42)

# Simulation parameters
n_simulations = 10_000
investment_per_startup = 1_000_000  # €1M each

# Startup return distributions (annual returns as decimals)
# Example: 0.25 = 25% return

# CloudAI: Normal(mean=25%, std=15%)
# GreenGrid: Normal(mean=18%, std=8%)
# HealthTrack: Normal(mean=30%, std=25%)
# FinFlow: Uniform(min=10%, max=35%)

print("Starter code loaded. Begin your analysis!")
print(f"Investment per startup: €{investment_per_startup:,}")
print(f"Number of simulations: {n_simulations:,}")
```

```
Starter code loaded. Begin your analysis!
Investment per startup: €1,000,000
Number of simulations: 10,000
```

Helpful Functions (Optional Use)

```
def calculate_portfolio_metrics(returns_1, returns_2, investment=1_000_000):
    """
    Calculate metrics for a portfolio of two startups

    Parameters:
    - returns_1: array of returns for startup 1 (as decimals, e.g., 0.25 for
25%)
    - returns_2: array of returns for startup 2
    - investment: amount invested in each startup

    Returns:
    - Dictionary with portfolio metrics
    """
    # Portfolio returns (equal weight)
    portfolio_returns = 0.5 * returns_1 + 0.5 * returns_2

    # Calculate absolute returns in euros
    absolute_returns = portfolio_returns * 2 * investment

    metrics = {
        'mean_return': portfolio_returns.mean(),
        'std_dev': portfolio_returns.std(),
        'prob_loss': (portfolio_returns < 0).mean(),
        'prob_high_return': (portfolio_returns > 0.5).mean(),
        'var_5': np.percentile(portfolio_returns, 5),
        'expected_profit': absolute_returns.mean()
    }

    return metrics

def expected_shortfall(returns, percentile=10):
    """
    Calculate the expected shortfall (conditional VaR)

    Parameters:
    - returns: array of returns
    - percentile: percentile for worst-case scenarios

    Returns:
    - Average return in the worst X% of scenarios
    """
    threshold = np.percentile(returns, percentile)
    worst_returns = returns[returns <= threshold]
    return worst_returns.mean() if len(worst_returns) > 0 else 0
```

```
# Example usage (delete or modify as needed):
print("Helper functions loaded and ready to use!")
```

```
Helper functions loaded and ready to use!
```

Submission Requirements

Required Deliverables

1. One-Slide Recommendation

Create one-slide (PDF) containing:

- **Recommendation:** Which 2 startups to invest in (clearly stated)
- **Expected Return:** Total expected return in € and %
- **Risk Assessment:** Your personal risk assessment
- **Justification:** 2-3 bullet points explaining why this pair
- **Visual:** One chart comparing the 6 portfolio options (optional but recommended)

2. Presentation Preparation

Be ready to present your recommendation in **3 minutes**:

- 1 minute: State recommendation and key metrics
- 1 minute: Explain your analysis approach
- 1 minute: Justify why this is the best choice for TechVenture

Tips for Success

Strategy Suggestions

1. **Start Simple:** Get one startup simulation working before scaling
2. **Verify Distributions:** Plot histograms to check your simulations look correct
3. **Think Like an Investor:** Consider both return AND risk
4. **Use Vectorization:** NumPy operations are faster than loops
5. **Document Assumptions:** Be clear about any choices you make

Common Pitfalls to Avoid

- Don’t forget to convert percentages to decimals (25% = 0.25)
- Remember you’re investing €2M total (€1M each in 2 startups)
- Check that your uniform distribution is implemented correctly
- Consider that high return often comes with high risk

Good Luck!

Bibliography